

BATTERY DISCHARGE ESTIMATION SPECIFICATION

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DETAILED DESCRIPTION

The magnitude of accumulated entropy generation until complete discharge (AEGD) is applied to rapidly estimate the remaining discharge time (RDT) of lithium-ion (Li-ion) batteries. This approach operates on real-time prediction of RDT during a single discharge cycle and is applicable across diverse operating conditions and battery types. Experimental validation tests are conducted on 18650 and 27000 Li-ion batteries with different capacities and discharge rates. Additional verification test results are presented using independent data from 14500 polymer Li-ion batteries. It is demonstrated that the method accurately identifies RDT for (i) variable operating conditions, (ii) from an arbitrary discharge voltage point, and (iii) fluctuating voltage profiles.

Li-ion batteries are the primary choice for applications ranging from small-scale portable instruments to large-scale electric vehicles. The choice is driven by their favorable characteristics, including high energy and power density, higher cycle life, reduced cost, low self-discharge, etc. Nevertheless, their operational performance and reliability in an application are greatly influenced by the remaining discharge time (RDT), which directly impacts the ability to sustain full discharge.

Accurate prediction of the RDT of lithium-ion batteries plays a critical role in optimizing their operational performance. An effective RDT model can provide valuable insights into the battery's available capacity, preventing issues such as thermal runaway, short circuit fault, overcharging and over-discharging, etc. These factors lead to unexpected shutdowns and potential damage and require substantial expenses for replacement and repairs due to irreversible damage. By safeguarding the battery's integrity through precise capacity forecasting, accurate RDT predictions can significantly extend the lifespan of these essential energy storage systems.